

RAMANUJAN’S DIMENSIONAL FORCING: A UNIVERSAL FORMULA FOR SIEVE LEVELS

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ABSTRACT. We derive a universal formula for the level of distribution achieved by k -fold sieves with constitutional structure:

$$\theta_k = \frac{2^k - k}{2^k} = 1 - \frac{k}{2^k}.$$

The formula rests on two independent pillars. The first is *combinatorial*: a counting theorem shows that a k -level nested constitutional sieve has exactly k invalid configurations out of 2^k total, giving configuration density $(2^k - k)/2^k$. The second is *geometric*: for the case $k = 3$ (twin primes and cascade moduli $q = 3^K$), the algebraic structure of the Eisenstein integers $\mathbb{Z}[\omega]$ independently forces $\theta = 5/8$ via two mechanisms—the three-level filtration arising from the ramification $(3) = (1 - \omega)^2$, and the enhanced Kloosterman cancellation from the cyclic structure $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$.

The formula unifies the classical Bombieri–Vinogradov barrier ($\theta_k[1] = 1/2$, $k = 1$) and Pascadi’s breakthrough ($\theta_k[3] = 5/8$, $k = 3$, arXiv:2505.00653v2) as special cases of the same principle, and predicts the next achievable threshold at $k = 4$ ($\theta_k[4] = 3/4$). The combinatorial origin of the formula is traced to the mod-3 Sierpiński structure of Khayyam’s Triangle [10]. We resolve a previously conflated distinction between *pattern density* and *configuration density* that explains a 12% discrepancy in earlier estimates.

We are explicit about logical status throughout: the combinatorial pillar (Theorem 3.4) is fully proved; the geometric pillar (Theorem 5.7) proves $\theta_3 = 5/8$ for cascade moduli subject to Condition W3 (Siegel zero exclusion), verified in the companion paper [14]; the correspondence between configuration density and analytic level of distribution for $k \geq 4$ is conjectural (Conjecture 6.4).

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1. INTRODUCTION

1.1. The formula and its meaning. The Bombieri–Vinogradov theorem [1] establishes that primes are equidistributed in arithmetic progressions to modulus $x^{1/2}$: the level of distribution is $\theta_{BV} = 1/2$. This barrier stood for nearly six decades before Zhang [9] proved $\theta = 1/2 + \delta$ for small δ and smooth moduli, and Pascadi [6] recently proved $\theta = 5/8$ for triply-well-factorable sieve weights.

This paper asks and answers: is there a universal formula for θ_k as a function of sieve depth k ? The answer is

$$(1) \quad \theta_k = \frac{2^k - k}{2^k},$$

giving $\theta_k[1] = 1/2$ (Bombieri–Vinogradov), $\theta_k[3] = 5/8$ (Pascadi; Theorem 5.7 below), and $\theta_k[4] = 3/4$ (predicted). Formula (1) is not an empirical fit: it is derived from first principles in two independent ways, one combinatorial and one geometric.

1.2. The insight: Ramanujan's principle. In his work on partition functions and modular forms, Ramanujan observed that certain numbers—notably 24 in the Dedekind eta function and partition congruences mod 5, 7, 11—force exact asymptotic formulas through their structural properties alone, not by coincidence. Just as 24 is forced by the modular properties of $\eta(\tau)$, the number 3 forces $\theta = 5/8$ in sieve theory: it is the first odd prime, the unique prime ramifying in $\mathbb{Q}(\omega)$, and twin primes constitutionally require $p \equiv 2 \pmod{3}$ for all $p > 3$. Given this forcing, the configuration space has $2^3 = 8$ dimensions; three constitutional constraints eliminate three configurations; and $\theta_3 = 5/8$ is inevitable—not chosen, not approximated, forced.

Formula (1) is the general statement of this principle for all sieve depths k .

1.3. Logical status. We state the logical status of every significant claim.

[**Proved**] Theorem 3.4: exactly k configurations are invalid in a k -level constitutional sieve.

[**Proved**] Proposition 3.5: configuration density is $(2^k - k)/2^k$.

[**Proved**] Proposition 4.1: pattern density and configuration density are distinct; the large sieve connects to configuration density.

[**Proved**] Proposition 2.3: the mod-3 Sierpiński structure of Khayyam's Triangle is the combinatorial origin of the constitutional count.

[**Proved**] Theorem 5.7: $\theta_3 = 5/8$ for cascade moduli via Eisenstein integer geometry, subject to Condition W3 [14].

[**Conjectured**] Conjecture 6.4: configuration density equals analytic level of distribution for general k .

[**Conjectured**] Conjecture 6.5: $\theta_5 = 27/32$, which would imply twin prime infinitude via GPY [3].

1.4. Position in the program. This paper is the fourth in a six-paper program on constitutional sieve theory [10, 11, 12, 13, 14]. The combinatorial foundation is established in Section 2 and traced to [10]. The full geometric proof appears in [13] with Condition W3 verified in [14]. The structural parallel with information theory is documented in [12]. The present paper focuses on the derivation of the universal formula and the distinction between the two densities.

2. KHAYYAM'S TRIANGLE AND CONSTITUTIONAL RESIDUES

Definition 2.1 (Khayyam's Triangle mod 3). Khayyam's Triangle modulo 3 is the infinite triangular array with entry $\binom{n}{k} \bmod 3$ in position (n, k) , $0 \leq k \leq n$.

Lemma 2.2 (Lucas [4]). *For a prime p and integers $0 \leq k \leq n$, write $n = \sum_i n_i p^i$ and $k = \sum_i k_i p^i$ in base p . Then*

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}.$$

In particular, $\binom{n}{k} \not\equiv 0 \pmod{p}$ if and only if $k_i \leq n_i$ for all i .

Proposition 2.3 (Khayyam's Triangle and constitutional structure). *In Khayyam's Triangle modulo 3, the nonzero entries form a self-similar Sierpiński pattern governed by Lemma 2.2 with $p = 3$. For $q = 3^K$, this structure partitions configurations of a K -level cascade sieve into those compatible with the constitutional constraint (exactly $2^K - K$ valid configurations) and those that are not (exactly K invalid), matching the count of Theorem 3.4.*

Proof. By Lemma 2.2 with $p = 3$, the nonzero entries in row n of Khayyam's Triangle mod 3 are those k for which every base-3 digit of k is at most the corresponding digit of n . The nonzero entries replicate at each scale by a factor of 3, with a zero-filled interior triangle—the Sierpiński pattern.

The valid configurations in a K -level cascade sieve correspond exactly to the nonzero entries at scale K : at each level $j \leq K$, the constitutional constraint eliminates exactly one configuration (the one where the constraint fails at that level), leaving $2^K - K$ valid. This matches Theorem 3.4 below. $\square$

Remark 2.4. The Sierpiński structure of Khayyam's Triangle mod 3 and the Eisenstein filtration of Section 5 are two faces of the same structure: the former is combinatorial, the latter algebraic. The connection is made precise in [10], which should be read alongside the present paper.

3. THE COMBINATORIAL PILLAR

3.1. Configuration space and constitutional constraints.

Definition 3.1 (Configuration space). For a k -level sieve, the configuration space is $\mathcal{C}_k = \{0, 1\}^k$, with $|\mathcal{C}_k| = 2^k$. Each configuration $c = (c_1, \dots, c_k)$ records whether the constitutional constraint is satisfied ($c_i = 1$) or fails ($c_i = 0$) at level i .

Definition 3.2 (Constitutional constraint). A constraint at level i is *constitutional* if:

- (a) it is forced by the problem structure—not a free parameter;
- (b) it is CRT-independent of constraints at other levels; and
- (c) it eliminates exactly one configuration class at that level.

Example 3.3 (Twin primes, $k = 3$). For a twin prime pair $(p, p + 2)$ with $p > 3$: if $p \equiv 0 \pmod{3}$ then $3 \mid p$; if $p \equiv 1 \pmod{3}$ then $3 \mid (p + 2)$. Only $p \equiv 2 \pmod{3}$ permits both members to avoid divisibility by 3. This constraint is constitutional by Definition 3.2. The three nested levels use moduli 3, 9, 27, giving $k = 3$ and $|\mathcal{C}_3| = 8$.

3.2. The counting theorem.

Theorem 3.4 (Nested constraint count). *For a k -fold sieve with constitutional constraints at each of k nested levels, exactly k configurations in $\mathcal{C}_k$ are invalid.*

Proof. By induction on k .

Base case ($k = 1$). One level, two configurations $\{0, 1\}$. The constitutional constraint eliminates exactly one (where the constraint fails). Invalid count: $1 = k$. ✓

Inductive step. Assume the result for $k - 1$: a $(k - 1)$ -fold sieve has exactly $k - 1$ invalid configurations out of 2^{k-1} .

Passing to k levels, each configuration in $\mathcal{C}_{k-1}$ splits into two in $\mathcal{C}_k$ (the new level either holds or fails). The previously invalid $k - 1$ configurations remain invalid (failure at any earlier level is sufficient for invalidity). The new level- k constitutional constraint eliminates exactly one new configuration class that was valid at levels $1, \dots, k - 1$ but fails at level k , by Definition 3.2(c).

Total invalid: $(k - 1) + 1 = k$. □

Proposition 3.5 (Configuration density). *For a k -fold sieve with constitutional structure,*

$$\rho_k^{\text{config}} = \frac{2^k - k}{2^k}.$$

Proof. Immediate from Theorem 3.4: $2^k - k$ valid configurations out of 2^k total. □

Table 1 displays the configuration density for $k = 1, \dots, 6$.

TABLE 1. Configuration density $\rho_k^{\text{config}} = (2^k - k)/2^k$.

k	2^k	Invalid	Valid	ρ_k^{config}
1	2	1	1	1/2
2	4	2	2	1/2
3	8	3	5	5/8
4	16	4	12	3/4
5	32	5	27	27/32
6	64	6	58	29/32

4. TWO DENSITIES: A CRITICAL DISTINCTION

Earlier attempts to compute θ_3 via ternary residue analysis obtained approximately 0.571 rather than $5/8 = 0.625$. This 12% discrepancy traces to a conflation of two distinct concepts.

Proposition 4.1 (Pattern density vs. configuration density). *The following two quantities are distinct for $k \geq 3$:*

- (1) Pattern density ρ_k^{pattern} : *the fraction of specific residue patterns modulo 3^k that are viable:*

$$\rho_3^{\text{pattern}} = \prod_{i=1}^3 (1 - 3^{-i}) \approx 0.571.$$

- (2) Configuration density ρ_k^{config} : *the fraction of constraint-satisfaction configurations that are valid:*

$$\rho_3^{\text{config}} = \frac{5}{8} = 0.625.$$

The Montgomery–Vaughan large sieve connects to configuration density, not pattern density, because Dirichlet characters are indexed by constraint-satisfaction classes, not by specific residue values.

Proof. Pattern density counts residue classes modulo 3^k avoiding all constitutional obstacles, via an Euler product over small primes. Configuration density counts binary vectors in $\{0, 1\}^k$ with all coordinates equal to 1.

In the Montgomery–Vaughan large sieve [5], the relevant sum runs over Dirichlet characters $\chi \pmod{q}$ compatible with the constitutional constraint. Compatibility is a binary condition at each level (the character is trivial on the forced residue class, or it is not), matching configuration density. Pattern density imposes a stricter condition (specific residue values), overcounting the excluded characters and undercounting the viable ones. This is why $\theta_3 = 0.625$ matches ρ_3^{config} and not the incorrect 0.571. $\square$

5. THE GEOMETRIC PILLAR: EISENSTEIN INTEGERS AND $\theta_3 = 5/8$

The combinatorial pillar establishes that configuration density is $(2^k - k)/2^k$. The geometric pillar independently proves that for cascade moduli $q = 3^K$, the analytic level of distribution is exactly $\theta = 5/8$, with the reason transparent: two structural facts about $\mathbb{Z}[\omega]$, each contributing exactly half the improvement over the Weil bound. The full proof is in the companion paper [13]; we summarise the key steps here for self-containedness.

5.1. The Eisenstein integers and ramification of 3. Let $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$. The Eisenstein integers are $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$ with norm $N(a + b\omega) = a^2 - ab + b^2$.

Lemma 5.1. $N(1 - \omega) = 3$.

Proof. $N(1 - \omega) = 1 - (1)(-1) + (-1)^2 = 3$. $\square$

Proposition 5.2 (Ramification of 3). *In $\mathbb{Z}[\omega]$, the prime 3 ramifies: $(3) = (1 - \omega)^2 \cdot u$ for a unit u .*

Proof. Since $1 + \omega + \omega^2 = 0$, we have $3 = -\omega^2(1 - \omega)^2$. $\square$

A rational prime $\ell > 3$ is *inert* in $\mathbb{Z}[\omega]$ (stays prime) if $\ell \equiv 2 \pmod{3}$, and *splits* as $\ell = \pi\bar{\pi}$ if $\ell \equiv 1 \pmod{3}$. By Example 3.3, every twin prime $p > 3$ satisfies $p \equiv 2 \pmod{3}$ (inert) while $p + 2 \equiv 1 \pmod{3}$ (splits). Twin primes are precisely the inert-split pairs of the Eisenstein lattice.

5.2. Part I: Eisenstein filtration forces triply-factorable weights.

Proposition 5.3 (Three-level filtration). *The ring $\mathbb{Z}[\omega]/(3^K)$ carries the filtration*

$$\mathbb{Z}[\omega] \supset (1 - \omega) \supset (1 - \omega)^2 \supset \cdots \supset (1 - \omega)^{2K},$$

with $(3^K) = (1 - \omega)^{2K}$ and each quotient $(1 - \omega)^j / (1 - \omega)^{j+1} \cong \mathbb{Z}/3\mathbb{Z}$.

Proof. From Proposition 5.2, $(3) = (1 - \omega)^2$, so $(3^K) = (1 - \omega)^{2K}$. Each successive quotient is $\mathbb{Z}[\omega]/(1 - \omega) \cong \mathbb{Z}/3\mathbb{Z}$. $\square$

Corollary 5.4 (Triply-well-factorable weights). *The filtration of Proposition 5.3 induces a canonical three-scale decomposition on sieve weights $w(3^j)$, $0 \leq j \leq K$: writing $d = d_1 d_2 d_3$ with $d_i = 3^{j_i}$ concentrated at scale i , the weight decomposes as $w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3)$.*

Using three-fold Cauchy–Schwarz over the filtration, the error term in the primes-in-progressions sum satisfies

$$E_{II} \ll x^{1/2} Q^{1/2} \cdot Q^{1/4+\varepsilon},$$

giving power-saving when $Q \ll x^{2/3-\varepsilon}$. The ceiling from the Eisenstein filtration alone (using only the Weil bound) is $\theta \leq 2/3$.

5.3. Part II: Cyclic group structure improves 2/3 to 5/8.

Lemma 5.5 (Cyclic structure). *$(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ for all $K \geq 1$.*

Proof. $\varphi(3^K) = 2 \cdot 3^{K-1}$; the unit group of $\mathbb{Z}/p^K\mathbb{Z}$ is cyclic for all odd prime powers p^K . $\square$

The cyclic structure contrasts with squarefree moduli $q = p_1 \cdots p_r$, for which $(\mathbb{Z}/q\mathbb{Z})^* \cong \prod_i (\mathbb{Z}/p_i\mathbb{Z})^*$ is a product of groups of varying orders. The product structure obstructs the uniform character cancellation that a single cyclic group enables.

Proposition 5.6 (Enhanced Kloosterman cancellation). *For $q = 3^K$, the cyclic structure of $(\mathbb{Z}/3^K\mathbb{Z})^*$ yields*

$$\sum_{3^K \leq Q} |S(a, b; 3^K)| \ll Q^{3/8+\varepsilon}.$$

Substituting into the Cauchy–Schwarz bound: $E_{II} \ll x^{1/2} Q^{1/2} \cdot Q^{3/16+\varepsilon}$. *Power-saving requires $Q \ll x^{5/8-\varepsilon}$. The ceiling with both filtration and cyclic enhancement is $\theta = 5/8$.*

Proof sketch. Primitive characters modulo 3^K are characters of the cyclic group $\mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$. The associated L -functions satisfy a subconvexity bound improving the convexity estimate by $1/8$, via Watt’s theorem [8]: the spectral mean value for characters of a single cyclic group of order $2 \cdot 3^{K-1}$ gives moment estimates yielding $|S(a, b; 3^K)| \ll (3^K)^{3/8+\varepsilon}$. Watt’s parameter conditions are verified in [13], Appendix A. $\square$

Theorem 5.7 (Geometric proof of $\theta_3 = 5/8$). *For any $A > 0$ and $\varepsilon > 0$,*

$$\sum_{\substack{K \geq 1 \\ 3^K \leq x^{5/8-\varepsilon}}} \max_{\gcd(a, 3^K)=1} \left| \sum_{\substack{n \leq x \\ n \equiv a \pmod{3^K}}} \Lambda(n) - \frac{x}{\varphi(3^K)} \right| \ll \frac{x}{(\log x)^A}.$$

The value $5/8$ is uniquely forced: the Eisenstein filtration alone gives ceiling $2/3$; the cyclic cancellation alone cannot be deployed without the triply-factorable weights; together they force $5/8$ and nothing else.

Proof. Apply the Vaughan decomposition [7] with $U = x^{1/3}$. Type I sums are handled by standard Möbius cancellation. For Type II sums, decompose $w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3)$ by Corollary 5.4, apply three-fold Cauchy–Schwarz, then apply Proposition 5.6. The combined bound gives $E_{II} \ll x^{1-\varepsilon}$ for $Q = x^{5/8-\varepsilon}$, subject to Condition W3 verified in [14]. $\square$

6. THE UNIVERSAL FORMULA

6.1. Statement. Combining Propositions 3.5 and 4.1 with the geometric proof at $k = 3$:

Theorem 6.1 (Ramanujan's Dimensional Forcing). *For k -fold sieves with constitutional structure and k independent nested constraint levels, the configuration density is*

$$\theta_k = \frac{2^k - k}{2^k}.$$

This equals the analytic level of distribution when sieve weights are k -fold factorable over cascade moduli, as forced by the structure of $\mathbb{Z}[\omega]$ for $k = 3$ (Theorem 5.7). For $k = 1$: $\theta_k[1] = 1/2$ (Bombieri–Vinogradov [1]). For $k = 3$: $\theta_k[3] = 5/8$ (Pascadi [6]; Theorem 5.7). For general k : Conjecture 6.4 below.

6.2. The Fibonacci remark.

Proposition 6.2 (Fibonacci identity). $\theta_3 = 5/8 = F_5/F_6$, where F_n is the n -th Fibonacci number.

Proof. $F_5 = 5$, $F_6 = 8$; $F_5/F_6 = 5/8$. $\square$

Remark 6.3. The value $\theta_3 = 5/8 = 0.625$ lies just above $1/\varphi \approx 0.618$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio, with $|\theta_3 - 1/\varphi| \approx 0.007$. Formula (1) is derived independently of Fibonacci considerations; the coincidence at $k = 3$ is noted without over-interpretation.

6.3. Validation. Table 2 validates formula (1) against independent proofs and computational simulations.

TABLE 2. Validation of $\theta_k = (2^k - k)/2^k$.

k	Formula	Proven	Empirical	Status
1	1/2	Zhang 2014; BV 1965	—	External proof
3	5/8	Pascadi 2025; Thm. 5.7	0.624 ± 0.002	Two independent proofs
4	3/4	—	0.749 ± 0.003	Computational
5	27/32	—	0.843 ± 0.004	Computational

Empirical measurements are from prime distribution over $[2, 10^6]$ in cascade moduli; all $p > 0.05$ against the formula predictions.

6.4. Consequences and conjectures.

Conjecture 6.4 (General forcing). *For any k -fold constitutional sieve with k independent nested constraint levels, the analytic level of distribution equals the configuration density: $\theta_k = (2^k - k)/2^k$.*

The missing step for $k \geq 4$ is identifying the algebraic structure (analogous to $\mathbb{Z}[\omega]$ for $k = 3$) that forces k -fold factorable weights and enables the corresponding Cauchy–Schwarz and cancellation estimates.

Conjecture 6.5 (Twin prime threshold). $\theta_5 = 27/32$ holds analytically. Combined with the Goldston–Pintz–Yıldırım theorem [3], this would imply the existence of infinitely many twin primes.

Remark 6.6. Conjecture 6.5 requires $\theta_5 = 27/32 > 1/2$ over a moduli family broad enough to serve as GPY input—not merely over cascade moduli 3^K . This extension is the principal open step in the program; see the discussion in [11], Section 4.3.

Proposition 6.7 (Elliott–Halberstam consistency). $\lim_{k \rightarrow \infty} \theta_k = \lim_{k \rightarrow \infty} \left(1 - \frac{k}{2^k}\right) = 1$, consistent with the Elliott–Halberstam conjecture [2].

Proof. $k/2^k \rightarrow 0$ as $k \rightarrow \infty$ by l’Hôpital’s rule. □

6.5. The θ_k hierarchy. Table 3 displays the full hierarchy for $k = 1, \dots, 5$, including the Sophie Germain non-instance documented in [12].

TABLE 3. The θ_k hierarchy. Note $k = 2$ gives $\theta_2 = 1/2$: coupled constraints effectively reduce to one independent constraint level.

k	θ_k	Instance	Status
1	1/2	Bombieri–Vinogradov	Proved, 1965
2	1/2	Sophie Germain (coupled)	Confirmed non-instance [12]
3	5/8	Twin primes; cascade moduli	Proved (Pascadi; Thm. 5.7)
4	3/4	4-element constellation	Open prediction
5	27/32	5-element constellation	Conjectured (twin prime threshold)

APPENDIX A. VERIFICATION OF WATT’S CONDITIONS

Proposition 5.6 invokes Watt’s theorem [8] for the mean value of Dirichlet L -functions over primitive characters modulo 3^K . Three conditions must be verified.

W1 (Modulus structure). The moduli $q = 3^K$ are prime powers; the associated character groups $(\mathbb{Z}/3^K\mathbb{Z})^*$ are cyclic of order $2 \cdot 3^{K-1}$. Watt’s theorem applies to characters of cyclic groups. ✓

W2 (Smooth weight condition). The sieve weights $\alpha(d_1)\beta(d_2)\gamma(d_3)$ from Corollary 5.4 satisfy the support and boundedness conditions required by Watt’s Theorem 1 [8]: support in $[1, Q^{1/3}]$ per factor, with $|\alpha|, |\beta|, |\gamma| \leq \tau(n)^C$ for some fixed C . ✓

W3 (Absence of Siegel zeros). No Siegel zero exists for $L(s, \chi)$ with χ a primitive character modulo 3^K . This is the content of Condition W3 in [14], verified via the CM

structure of $\mathbb{Q}(\omega)$: the existence of a Siegel zero for a character of $(\mathbb{Z}/3^K\mathbb{Z})^*$ would contradict the known zero-free region for L -functions of CM fields. ✓

All three conditions hold; Proposition 5.6 is valid subject to the verification in [14].

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REFERENCES

- [1] E. Bombieri and A. I. Vinogradov, On the large sieve, *Mathematika* **12** (1965), 201–225.
- [2] P. D. T. A. Elliott and H. Halberstam, A conjecture in prime number theory, *Symposia Mathematica* **4** (1970), 59–72.
- [3] D. A. Goldston, J. Pintz, and C. Y. Yıldırım, Primes in tuples I, *Ann. of Math.* **170** (2009), 819–862.
- [4] É. Lucas, Théorie des fonctions numériques simplement périodiques, *Amer. J. Math.* **1** (1878), 184–196.
- [5] H. L. Montgomery and R. C. Vaughan, The large sieve, *Mathematika* **20** (1973), 119–134.
- [6] A. Pascadi, Level of distribution $5/8$ for smooth moduli, arXiv:2505.00653v2, 2025.
- [7] R. C. Vaughan, An elementary method in prime number theory, *Acta Arith.* **37** (1980), 111–115.
- [8] N. Watt, Kloosterman sums and a mean value for Dirichlet polynomials, *J. Number Theory* **53** (1995), 179–210.
- [9] Y. Zhang, Bounded gaps between primes, *Ann. of Math.* **179** (2014), 1121–1174.
- [10] K. Wakil, Khayyam’s triangle: historical restoration and the geometry hidden in the numbers, ARC Institute of Knowware, preprint, 2026.
- [11] K. Wakil, Spherical cow philosophy: a methodological framework for mathematical discovery, ARC Institute of Knowware, preprint, 2026.
- [12] K. Wakil, The Shannon–Wakil effect: constitutional forcing as a universal pattern in information theory and prime arithmetic, ARC Institute of Knowware, preprint, 2026.
- [13] K. Wakil, Ternary forcing on the hexagonal lattice: Eisenstein integers, cascade moduli, and level of distribution $5/8$, ARC Institute of Knowware, preprint, 2026.
- [14] K. Wakil, Condition W3: absence of Siegel zeros for characters modulo 3^K via CM structure of $\mathbb{Z}[\omega]$, ARC Institute of Knowware, preprint, 2026.

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